# WEEK 8:Database Relationship Analysis using joins

**Scenario:**

An e-commerce company selling ceramic products maintains multiple tables such as **CUSTOMER**, **ORDERS**, **PRODUCT**, **ORDER_DETAILS**, and **SUPPLIER**. To generate meaningful business reports, the company needs to analyze relationships between these tables using SQL joins.

**INNER JOIN**

**1.Display customer details along with their orders**

```
SQL> SELECT C.CUSTOMER_NAME,O.ORDER_ID,O.TOTAL_AMOUNT
  2  FROM CUSTOMER C INNER JOIN ORDERS O
  3  ON C.CUSTOMER_ID=O.CUSTOMER_ID;

CUSTOMER_NAME                ORDER_ID TOTAL_AMOUNT
-------------------------- ---------- ------------
Sweety                           1001          550
Arun                             1002          250
Priya                            1003          600
Kumar                            1004          200
Meena                            1005         1200
Ravi                             1006          750

6 rows selected.
```

**2.Display order details along with payment information**

```
SQL> SELECT O.ORDER_ID,O.TOTAL_AMOUNT,O.ORDER_DATE,O.CUSTOMER_ID,P.PAYMENT_DATE,P.TRANSACTION_AMOUNT,P
.PAYMENT_MODE
  2  FROM ORDERS O INNER JOIN PAYMENT P
  3  ON O.ORDER_ID=P.ORDER_ID;

  ORDER_ID TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID PAYMENT_D TRANSACTION_AMOUNT
---------- ------------ --------- ----------- --------- ------------------
PAYMENT_MODE
--------------------
      1001          550 28-SEP-26         101 26-SEP-26                550
UPI

      1002          250 27-AUG-26         102 27-AUG-26                250
Credit Card

      1003          600 09-SEP-26         103 09-OCT-26                600
Cash


  ORDER_ID TOTAL_AMOUNT ORDER_DAT CUSTOMER_ID PAYMENT_D TRANSACTION_AMOUNT
---------- ------------ --------- ----------- --------- ------------------
PAYMENT_MODE
--------------------
      1004          200 15-SEP-26         104 15-OCT-26                200
UPI

      1005         1200 29-SEP-26         105 29-OCT-26               1200
Debit Card

      1006          750 30-SEP-26         106 26-OCT-26                750
Net Banking


6 rows selected.
```

**3.Display customer without any purchase**

```
SQL> SELECT C.CUSTOMER_ID,C.CUSTOMER_NAME,C.MOBILE_NUMBER
  2  FROM CUSTOMER C
  3  LEFT JOIN ORDERS O
  4  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  WHERE O.ORDER_ID IS NULL
  6  ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME              MOBILE_NUMBER
----------- -------------------------- -------------
        111 Vihan                        9876541230
```

**4.Display product that has no sales**

```
SQL> SELECT P.PRODUCT_ID,P.PRODUCT_NAME,P.PRICE
  2  FROM PRODUCT P LEFT JOIN ORDER_DETAILS OD
  3  ON P.PRODUCT_ID = OD.PRODUCT_ID
  4  WHERE OD.PRODUCT_ID IS NULL
  5  ORDER BY P.PRODUCT_ID;

PRODUCT_ID PRODUCT_NAME         PRICE
---------- ---------------- ----------
P00012     Olive Tray             650
```

**5.Generate complete order details**

```
SQL> SELECT C.CUSTOMER_NAME,P.PRODUCT_NAME,OD.QUANTITY,O.ORDER_DATE,O.TOTAL_AMOUNT,PY.PAYMENT_STATUS
  2  FROM CUSTOMER C JOIN ORDERS O
  3  ON C.CUSTOMER_ID=O.CUSTOMER_ID
  4  JOIN ORDER_DETAILS OD
  5  ON O.ORDER_ID = OD.ORDER_ID
  6  JOIN PRODUCT P
  7  ON OD.PRODUCT_ID=P.PRODUCT_ID
  8  LEFT JOIN PAYMENT PY
  9  ON O.ORDER_ID=PY.ORDER_ID
 10  ORDER BY O.ORDER_DATE,C.CUSTOMER_ID;

CUSTOMER_NAME             PRODUCT_NAME      QUANTITY ORDER_DAT TOTAL_AMOUNT
------------------------- --------------- ---------- --------- ------------
PAYMENT_STATUS
-------------------------
Arun                      Wall hanging             4 27-AUG-26          250
 Success

Priya                     Coffee Cup               4 09-SEP-26          600
Success

Kumar                     Ceramic bowl             2 15-SEP-26          200
Success


CUSTOMER_NAME             PRODUCT_NAME      QUANTITY ORDER_DAT TOTAL_AMOUNT
------------------------- --------------- ---------- --------- ------------
PAYMENT_STATUS
-------------------------
Sweety                    Flower_Vase              2 28-SEP-26          550
Success

Meena                     Plant Pot                1 29-SEP-26         1200
Success

Ravi                      Plant Pot                1 30-SEP-26          750
Success


6 rows selected.
```

**6.Number of orders placed by each customer**

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  ORDER BY NUMBER_OF_ORDERS DESC;

CUSTOMER_ID CUSTOMER_NAME              NUMBER_OF_ORDERS
----------- ------------------------- ----------------
        105 Meena                                    2
        102 Arun                                     1
        103 Priya                                    1
        104 Kumar                                    1
        101 Sweety                                   1
        106 Ravi                                     1
        107 Ananya                                   1
        108 Snaha                                    1
        109 Rahul Verma                              1
        110 Karthik                                  1
        111 Vihan                                    0

11 rows selected.
```

## RIGHT JOIN

### 1.Dusplay all orders with customer name

```
SQL> SELECT O.ORDER_ID,O.ORDER_DATE,O.TOTAL_AMOUNT,O.ORDER_STATUS,C.CUSTOMER_ID,C.CUSTOMER_NAME
  2  FROM CUSTOMER C RIGHT JOIN ORDERS O
  3  ON  C.CUSTOMER_ID=O.CUSTOMER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS             CUSTOMER_ID CUSTOMER_NAME
---------- --------- ------------ ------------------------ ----------- -------------------------
      1001 28-SEP-26          550 Processing                       101 Sweety
      1002 27-AUG-26          250 Dispatched                       102 Arun
      1003 09-SEP-26          600 Dispatched                       103 Priya
      1004 15-SEP-26          200 Dispatched                       104 Kumar
      1005 29-SEP-26         1200 Processing                       105 Meena
      1006 30-SEP-26          750 Processing                       106 Ravi

6 rows selected.
```

## 2.Display all payment records with order details

```
SQL> SELECT P.*,O.ORDER_ID,O.ORDER_DATE,ORDER_STATUS
  2  FROM PAYMENT P RIGHT JOIN ORDERS O
  3  ON P.ORDER_ID = O.ORDER_ID
  4  ORDER BY ORDER_DATE;

PAYMENT_ID   ORDER_ID PRODUCT_ID   PRODUCT_NAME         PAYMENT_D PAYMENT_MODE         PAYMENT_STATUS
TRANSACTION_AMOUNT   ORDER_ID ORDER_DAT ORDER_STATUS
---------- ---------- ---------- -------------------- --------- -------------------- ---------------
---------- ---------- --------- ------------------------
         2       1002 P00001       Ceramic Bowl         27-AUG-26 Credit Card           Success      2
50        1002 27-AUG-26 Dispatched
         3       1003 P00002       Coffee Cup           09-OCT-26 Cash                 Success       6
00        1003 09-SEP-26 Dispatched
         4       1004 P00006       Ceramic Plate        15-OCT-26 UPI                  Success       2
00        1004 15-SEP-26 Dispatched
         1       1001 P00004       Flower Vase          26-SEP-26 UPI                  Success       5
50        1001 28-SEP-26 Processing
         5       1005 P00003       Plant Pot            29-OCT-26 Debit Card           Success
      1200        1005 29-SEP-26 Processing
         6       1006 P00005       Wall Hanging         26-OCT-26 Net Banking          Success       7
50        1006 30-SEP-26 Processing

6 rows selected.
```

## 3.Display all payment records with order detail

```
SQL> SELECT PY.*,O.*
  2  FROM ORDERS O RIGHT JOIN PAYMENT PY
  3  ON O.ORDER_ID=PY.ORDER_ID
  4  ORDER BY PY.PAYMENT_DATE;

PAYMENT_ID   ORDER_ID PRODUCT_ID PRODUCT_NAME    PAYMENT_D PAYMENT_MODE
---------- ---------- ---------- --------------- --------- --------------------
PAYMENT_STATUS            TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID ORDER_DAT
------------------------- ------------------ ---------- ----------- ---------
TOTAL_AMOUNT ORDER_STATUS              CUS CUSTOMER_NAME
------------ ------------------------- --- -------------------------
PRODUCT_NAME                   QUANTITY
------------------------------ ----------
         2       1002 P00001     Ceramic Bowl    27-AUG-26 Credit Card
 Success                                 250       1002         102 27-AUG-26
         250 Dispatched                NO  Arun
Ceramic Bowl                            2


PAYMENT_ID   ORDER_ID PRODUCT_ID PRODUCT_NAME    PAYMENT_D PAYMENT_MODE
---------- ---------- ---------- --------------- --------- --------------------
PAYMENT_STATUS            TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID ORDER_DAT
------------------------- ------------------ ---------- ----------- ---------
TOTAL_AMOUNT ORDER_STATUS              CUS CUSTOMER_NAME
------------ ------------------------- --- -------------------------
PRODUCT_NAME                   QUANTITY
------------------------------ ----------
         1       1001 P00004     Flower Vase     26-SEP-26 UPI
Success                                 550       1001         101 28-SEP-26
         550 Processing                YES Sweety
Flower Vase                             2
```

```
PAYMENT_ID   ORDER_ID PRODUCT_ID PRODUCT_NAME    PAYMENT_D PAYMENT_MODE
---------- ---------- ---------- --------------- --------- --------------------
PAYMENT_STATUS            TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID ORDER_DAT
------------------------- ------------------ ---------- ----------- ---------
TOTAL_AMOUNT ORDER_STATUS              CUS CUSTOMER_NAME
------------ ------------------------- --- -------------------------
PRODUCT_NAME                    QUANTITY
------------------------------ ----------
         3       1003 P00002     Coffee Cup      09-OCT-26 Cash
Success                                 600       1003         103 09-SEP-26
         600 Dispatched                YES Priya
Coffee Mug                               4


PAYMENT_ID   ORDER_ID PRODUCT_ID PRODUCT_NAME    PAYMENT_D PAYMENT_MODE
---------- ---------- ---------- --------------- --------- --------------------
PAYMENT_STATUS            TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID ORDER_DAT
------------------------- ------------------ ---------- ----------- ---------
TOTAL_AMOUNT ORDER_STATUS              CUS CUSTOMER_NAME
------------ ------------------------- --- -------------------------
PRODUCT_NAME                    QUANTITY
------------------------------ ----------
         4       1004 P00006     Ceramic Plate   15-OCT-26 UPI
Success                                 200       1004         104 15-SEP-26
         150 Dispatched                YES Kumar
Ceramic Plate                            2


PAYMENT_ID   ORDER_ID PRODUCT_ID PRODUCT_NAME    PAYMENT_D PAYMENT_MODE
---------- ---------- ---------- --------------- --------- --------------------
PAYMENT_STATUS            TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID ORDER_DAT
------------------------- ------------------ ---------- ----------- ---------
TOTAL_AMOUNT ORDER_STATUS              CUS CUSTOMER_NAME
------------ ------------------------- --- -------------------------
PRODUCT_NAME                    QUANTITY
------------------------------ ----------
         6       1006 P00005     Wall Hanging    26-OCT-26 Net Banking
Success                                 750       1006         106 30-SEP-26
         125 Processing                YES Ravi
Wall hanging                             1
```

```
PAYMENT_ID   ORDER_ID PRODUCT_ID PRODUCT_NAME    PAYMENT_D PAYMENT_MODE
---------- ---------- ---------- --------------- --------- --------------------
PAYMENT_STATUS            TRANSACTION_AMOUNT   ORDER_ID CUSTOMER_ID ORDER_DAT
------------------------- ------------------ ---------- ----------- ---------
TOTAL_AMOUNT ORDER_STATUS              CUS CUSTOMER_NAME
------------ ------------------------- --- -------------------------
PRODUCT_NAME                    QUANTITY
------------------------------ ----------
         5       1005 P00003     Plant Pot       29-OCT-26 Debit Card
Success                                1200       1005         105 29-SEP-26
         150 Processing                NO  Meena
Plant Pot                                1


6 rows selected.
```

## JOINS

### 1.All products purchased by a customer

```
SQL> SELECT C.CUSTOMER_NAME,O.ORDER_ID,O.ORDER_DATE,P.PRODUCT_NAME,OD.QUANTITY,OD.TOTAL_PRICE
  2  FROM CUSTOMER C JOIN ORDERS O
  3  ON C.CUSTOMER_ID=O.CUSTOMER_ID
  4  JOIN ORDER_DETAILS OD
  5  ON O.ORDER_ID=OD.ORDER_ID
  6  JOIN PRODUCT P
  7  ON OD.PRODUCT_ID=P.PRODUCT_ID
  8  WHERE C.CUSTOMER_ID=105
  9  ORDER BY ORDER_DATE;

CUSTOMER_NAME              ORDER_ID ORDER_DAT PRODUCT_NAME      QUANTITY
------------------------- ---------- --------- --------------- ----------
TOTAL_PRICE
-----------
Meena                           1011 26-JUL-26 Book Mark               4
       400

Meena                           1005 29-SEP-26 Plant Pot               1
       150
```

### 2.Customer order report with order status

```
SQL> SELECT C.CUSTOMER_NAME,O.ORDER_ID,O.ORDER_DATE,O.ORDER_STATUS
  2  FROM CUSTOMER C JOIN ORDERS O
  3  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  4  ORDER BY O.ORDER_DATE;

CUSTOMER_NAME              ORDER_ID ORDER_DAT ORDER_STATUS
------------------------- ---------- --------- -------------------------
Meena                           1011 26-JUL-26 Dispatched
Arun                            1002 27-AUG-26 Dispatched
Priya                           1003 09-SEP-26 Dispatched
Kumar                           1004 15-SEP-26 Dispatched
Rahul Verma                     1009 18-SEP-26 Dispatched
Karthik                         1010 26-SEP-26 Dispatched
Sweety                          1001 28-SEP-26 Processing
Meena                           1005 29-SEP-26 Processing
Ravi                            1006 30-SEP-26 Processing
Ananya                          1007 18-OCT-26  Processing
Snaha                           1008 18-OCT-26 Dispatched

11 rows selected.
```

**3.Total amount spend by each customer**

```
SQL> SELECT P.PRODUCT_ID,
  2         P.PRODUCT_NAME,
  3         SUM(OD.QUANTITY) AS SOLD_QUANTITY,
  4         SUM(OD.TOTAL_PRICE) AS TOTAL_REVENUE
  5  FROM PRODUCT P
  6  JOIN ORDER_DETAILS OD
  7  ON P.PRODUCT_ID = OD.PRODUCT_ID
  8  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  9  ORDER BY TOTAL_REVENUE DESC;

PRODUCT_ID PRODUCT_NAME    SOLD_QUANTITY TOTAL_REVENUE
---------- --------------- ------------- -------------
P00009     Keychain                   10          1500
P00008     Spoons                      4          1000
P00007     Chip_Dip boul               2          1000
P00002     Coffee Mug                  4           600
P00010     Engraved Plates             2           580
P00004     Flower_Vase                 2           550
P00011     Book Mark                   4           400
P00006     Ceramic Plate               2           300
P00001     Ceramic bowl                2           250
P00003     Plant Pot                   1           150
P00005     Wall hanging                1           125

11 rows selected.
```

**4.Customer maximum orders**

```
SQL> SELECT C.CUSTOMER_ID,
  2         C.CUSTOMER_NAME,
  3         COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  HAVING COUNT(O.ORDER_ID) = (
  9      SELECT MAX(ORDER_COUNT)
 10      FROM (
 11          SELECT COUNT(*) AS ORDER_COUNT
 12          FROM ORDERS
 13          GROUP BY CUSTOMER_ID
 14      )
 15  );

CUSTOMER_ID CUSTOMER_NAME              NUMBER_OF_ORDERS
----------- ------------------------- ----------------
        105 Meena                                    2
```